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AMENDMENT 1
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MILITARY SPECIFICATION

INDICATORS, PRESSURE, HELICOPTER, VOLTAGE RATIO TYPE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for voltage ratio type pressure indicators that operate on alternating current and are designed for use in helicopters.

2 APPLICABLE DOCUMENTS

2.1 Government Documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-58097	Transmitters, Pressure, Helicopter, Voltage Ratio Type, General Specification for
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts: Packaging and Packing of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Aviation Research and Development Command, ATTN: DRDAV-ERS, 4300 Goodfellow Boulevard, St. Louis, MO 63120, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33638	Case, Instrument, Flange-Mounted, Aircraft
MS33639	Case, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Type of Testing. The type of testing to be accomplished on these indicators shall be as specified in the detail specification.

3.2 Selection of Specifications and Standards. Specifications and standards for required commodities or services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Nonmagnetic Materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.2 Nonferrous Materials. Nonferrous materials, shall be used for all parts of the indicator except where ferrous materials are essential.

3.3.3 Metals. Metals shall be of a corrosion-resistant type or shall be suitably treated to resist corrosion due to fuels, salt spray, and atmospheric or operating conditions either in storage or normal service.

3.3.4 Dissimilar Metals. Dissimilar metals, as defined in MS33586, shall not be used in intimate contact unless suitably protected against electrolytic corrosion.

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3.3.5 Toxic and Corrosive Fumes. The materials used in the indicators shall not, when burned, release fumes (or gases) which can combine with the atmosphere to produce toxic or corrosive substances.

3.3.6 Protective Treatment. Materials used in the indicator that are subject to deterioration when exposed to climatic or environmental conditions of service usage shall be protected against such deterioration. Use of protective coatings that will crack, chip, or scale with age or exposure to climatic or environmental conditions shall be avoided.

3.3.7 Recycled, Virgin and Reclaimed Materials. This specification shall require the use of reclaimed materials to the maximum extent possible (see 6.4).

3.4 Design and Construction. The indicator shall provide remote indication of aircraft functions when connected to a transmitter conforming to MIL-T-58097. It shall be of a voltage ratio type which operates a pointer that continuously indicates the measured pressure on a dial.

3.4.1 Power Supply. The indicator shall operate from a 26 $\pm$ 0.5V 400 $\pm$ 10 CPS single-phase power source. The indicator shall not be damaged by a voltage of 10 percent greater than maximum value specified.

3.4.1.1 Wiring. External wiring of the indicator shall be in accordance with Figure 1. Internal wiring shall be color coded and shall be connected as shown on Figure 2.

3.4.1.2 Electrical Connectors. Electrical connectors shall be in accordance with MIL-C-5015. Connector shells shall be protected against corrosion and shall meet the requirements specified herein. The connector polarized key shall be located within $\pm$ 20 degrees of the vertical centerline of side nearest top of indicator. Each hermetically sealed indicator shall be provided with an MS33678-10SL-3P receptacle.

3.4.2 Case. Unless otherwise specified in the detail specification, the case shall consist of a body and a bezel ring. The case shall be sufficiently rugged so that small distortions will not affect accuracy. A separate cap may be used on the back of the case for installing the electrical receptacle. Cap shall be of nonferrous low-density metal.

3.4.2.1 Bezel Ring. The bezel ring shall be made of nonferrous low density metal with a durable dull black finish. It shall be constructed so that it will properly position the cover glass.

3.4.2.2 Filler Tube. A tube (or other facility) for evacuating and filling the case shall be provided at the rear of case. It shall be beneath the level of back plate surface and be of such design to permit at least three re-fillings.

3.4.3 Cover Glass. The cover glass shall be clear, flat, and free from flaws that could interfere with normal reading of the instrument. Thickness of glass shall be the minimum practicable.

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3.4.3.1 Installation. Distance between outer face of cover glass and front edge of case shall not exceed 1/32 inch. Distance between inner face of cover glass and the indicator dial shall not exceed 0.140 inch.

3.4.4 Dial. The dial shall be in accordance with the detail specification. It shall be securely fastened to the front of the internal mechanism so that it will not loosen or turn when vibrated. Attaching devices shall be so located that they will not interfere with dial markings or prevent the markings from being properly positioned.

3.4.4.1 Markings. Numerals and letters on the dial shall conform to MS33558 and shall withstand usage encountered in normal service. Each numeral shall distinctly indicate the graduation to which it applies. Space between numerals shall not be less than 1/64 inch when used in a group.

3.4.4.2 Size. Dials installed in cases conforming to MS33639 shall have a minimum diameter across outside ends of the graduation markings of 1 3/4 inches. Dials installed in cases conforming to MS33638 shall have a minimum diameter across outside ends of graduation of 1 3/4 or 1 15/16 inches, as applicable.

3.4.5 Pointer. The pointer design shall be in accordance with the detail specification. In the event of power failure, the pointer shall move to a position below minimum scale point equivalent in degrees to approximately 3 percent of full scale.

3.4.6 Colors. Pointers, dials, and dial markings shall be finished in the color specified in the detail specifications.

3.4.7 Construction. The indicator shall be constructed so that no parts will come loose in normal service. Pivots, bearings, and gears shall be as nearly frictionless as practicable. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, handling, installation and service.

3.4.8 Filling Medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain no more than 0.006 milligram of water vapor per liter (dewpoint - 65°C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

3.5 Performance. The indicator shall operate as specified herein after being subjected to the following conditions:

a. Temperatures - temperatures ranging from -65° to +71°C.

b. Vibration - vibration of 5g's from 5 to 500 CPS.

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c. Temperature - altitude - pressure. Altitude of 25,000 feet at -55°C.

d. Thermal shock - eight cycles of alternate immersions in tap water at 85 ± 2°C and 5 ± 2°C.

e. Salt Fog - exposure to salt sea atmosphere for 48 hours without evidence of corrosion or deterioration.

3.5.1 Tolerances. The indicator shall operate within tolerances specified in the detail specification.

3.5.2 Dielectric Strength. The indicator shall withstand 200V rms at a commercial frequency for a period of 5 seconds.

3.5.3 Case Leakage. Case leakage shall not exceed 0.1 micron cubic foot per hour.

3.5.4 Fogging. The indicator shall be subjected to a temperature of 85° ± 5°C for a minimum period of 2 hours then the face of the cover glass shall be immediately immersed in an ice bath for 3 minutes. There shall be no evidence of moisture or oil fog inside the cover glass.

3.5.5 Damping. The time required for the indicator to respond to a step input equivalent to a 60 psi pressure change shall not exceed 2.0 seconds.

3.5.6 Electrical characteristics. The indicator shall have the following primary power electrical characteristics with a normal excitation of 26V, 400 cycle, single phase applied across terminals A and B.

3.5.6.1 Primary Circuit. With terminal C open circuited, the current drawn by the indicator shall not exceed 100 MA.

3.5.6.2 Power Input. The indicator power input shall not exceed 0.60w.

3.5.6.3 Open Circuit TAP Voltage. The open circuit TAP voltage and phase angle, with pointer restrained at the 0 percent, 50 percent, and 100 percent positions of full scale pressure, shall be measured using the circuit described in Figure 4. The TAP voltage and phase angle between indicator terminals A and C resulting from these measurements, shall be calculated and shall be within the following limits:

Percent Deflection	TAP Voltage Ratio	Phase Angle
0	.6060±.0075	1.5±0.5°
50	.5012±.0075	-0.5±0.8°
100	.4030±.0075	-3.5±1.1°

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3.5.6.4 Impedance. The Thevenin impedance of the indicator with the pointer restrained at the 0 percent, 50 percent, and 100 percent positions of full scale deflection shall be measured using the Maxwell bridge method (see Figure 5). The bridge input voltage shall be 26.0±0.05V, 400.0±0.1 HZ Sine wave with 0.5% maximum distortion. The impedance resulting from these measurements shall be calculated and shall be within the following limits:

Percent Deflection	Resistance(ohms)	Inductance (millihenries)
0	47.82±4.78	0.82±0.75
50	47.59±4.76	4.26±0.75
100	49.39±4.94	3.62±0.75

3.5.6.5 Power Supply Variations. The indicator shall meet the power supply variations of MIL-STD-704 (Category C) within tolerances specified in the detail specification.

3.5.7 Effect of Lead Length. When lead length between the indicator and transmitter is changed from 5 to 300 feet, the indicator shall not change more than 1° of arc.

3.5.8 Magnetic Effort. Compass deflection shall not exceed 1° of arc when the indicator is positioned 5 1/2 inches from the compass and horizontal magnetic field intensity as the indicator is 0.17 to 0.19 oersted.

3.6 Protective Finishes and Coatings.

3.6.1 Aluminum Alloy Parts. Aluminum alloy parts, unless contained in a hermetically sealed case, shall be coated with an anodic film conforming to MIL-A-8625.

3.6.1.1 Aluminum alloy parts that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541, except when abrasion resistance is a factor.

3.7 Screw Threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.8 Identification of Product. Indicators shall be marked for identification in accordance with MIL-STD-130. A name-plate containing identification information shall be permanently attached to rear of case by means other than glue or cement.

3.9 Part Numbering. All parts having the same manufacturer's part number shall be physically and functionally interchangeable. Item identification and part number requirements shall be governed by MIL-STD-100.

3.10 Workmanship.

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3.10.1 Dimensions and Tolerances. Dimensions and tolerances not specified shall be consistent with the best shop practices. Where dimensions and tolerances may affect interchangeability or performance of the indicator, they shall be held or limited accordingly.

3.10.2 Construction. The indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.10.3 The indicator shall be thoroughly cleaned. Burrs, sharp edges, and metal chips shall be removed. Loose, spattered or excess solder, and resin flash which might crumble, or other foreign material shall be removed both during and after assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements.

4.2 Classification of Tests. The inspection and testing of indicators shall be classified as acceptance tests and as specified in the detail specifications.

4.3 Test Conditions.

4.3.1 Standard Atmospheric Conditions. Wherever pressure and temperature for a test is not definitely specified, the test is to be made at atmospheric pressure (approx. 29.92 in. Hg) and at room temperature (approx. 25°C). Proper allowances must be made for differences from these standard conditions.

4.3.2 Attitude. The indicator shall be tested in accordance with the requirements specified in the detail specification.

4.3.3 Tapping. Unless otherwise specified, the indicator shall be lightly tapped or vibrated before each test reading is taken.

4.3.4 Test Voltage. The indicator shall be tested with an applied voltage of 26.0 ± 0.5V single-phase 400 ± 10 cycles A-C power.

4.3.5 Test Standard. A standard ratio transformer circuit conforming to Figure 3 shall be used for test of indicator.

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4.3.5.1 Test Standard Impedance. The Thevenin impedance of the Gertsch transformer, with associated external resistor and inductor connected to the tap, shall be measured using Maxwell bridge circuit shown in Figure 5. The bridge supply shall be 400. ± 0.5 HZ Sine wave with 0.5% maximum distortion. The impedances resulting from these measurements shall be calculated and shall be within the following limits:

Percent Deflection	Bridge Supply Voltage (Vrms)	Ratio Transformer Settings	Resistance (Ohms)	Inductance (Millihenry)
0	1.0	.5980	13.5±1.4	26.0±2.6
50	0.25	.4980	13.4±1.4	26.4±2.6
100	1.3	.4020	13.3±1.4	26.0±2.6

4.4 Acceptance Tests. Acceptance tests shall consist of:

a. Individual tests
b. Sampling tests
c. Manufacturing Run-in tests

4.4.1 Individual Tests. Each indicator shall be subjected to the following tests as described under 4.5:

a. Examination of product
b. Dielectric strength
c. Scale error and friction
d. Position error
e. Sealing
f. Damping
g. Pointer smoothness

4.4.2 Sampling Tests.

4.4.2.1 Lot. A lot shall consist of indicators manufactured under substantially the same conditions and submitted for acceptance at essentially the same time.

4.4.2.2 Sampling Plan A. One indicator shall be selected at random from each 100 or less. It shall be subjected to the following tests as described under 4.5:

a. Individual tests
b. Vibration
c. Low temperature
d. High temperature
e. Fogging

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4.4.2.3 Sampling Plan B. Unless otherwise specified, 3 indicators shall be selected at random from the first 15 produced on a contract or order and shall be subjected to the following tests as described under para 4.5:

a. Sampling Plan A tests
b. Electrical characteristics
c. Magnetic effect
d. Effect of lead length
e. Temperature-altitude
f. Low temperature exposure
g. High temperature exposure
h. Thermal shock
i. Salt fog
j. Shock

4.4.3 Rejection and Retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until extent and cause of failure have been determined. Individual tests may be continued pending investigation of the sampling test failure. After correction of failure has been made, all necessary tests shall be repeated. Final acceptance of a lot shall not be made until all indicators in the lot meet the specification.

4.4.4 Defects in Accepted Items. Investigation of a test failure could indicate that defects exist in items already accepted. When discovered, the contractor shall immediately advise the procuring activity of all defects likely to be found and proper methods of correcting them.

4.5 Test Methods.

4.5.1 Examination of Product. Each indicator shall be inspected for compliance to requirements herein pertaining to materials, design and construction, finish and workmanship. Prior to final assembly, dials and pointers shall be inspected as to proper dimensions, marking, and finishes.

4.5.2 Dielectric Strength. A potential of 200V rms at a commercial frequency shall be applied between connector pins and indicator case for 1 minute. There shall be no insulation breakdown as a result of this test. (If design of indicator is such that there exists between the test points a number of electrical circuit elements which form a conducting path for alternating current so that the indicator would be damaged by the test voltage, than those elements shall be removed, open circuited, or short circuited prior to application of the voltage.)

4.5.3 Scale Error and Friction. A standard transformer circuit shall be adjusted to provide voltages across terminals A and C corresponding to the pressure specified in the detail specification. The ratio transformer settings given in Table I shall be taken; the first before, and the second after the instrument is tapped. The difference between first and second readings shall not exceed friction tolerance specified in the detail specification. The second reading shall not exceed scale error specified in the detail specification. When

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indicator goes from power off to initial power on condition, the instrument should indicate an on scale reading at zero within the specified scale error tolerances. Scale error tests shall be conducted in both ascending and descending directions.

TABLE I

Percent Deflection	Ratio Transformer Setting
0	.5980
10	.5777
20	.5582
30	.5382
40	.5185
50	.4980
60	.4793
70	.4600
80	.4408
90	.4214
100	.4020

4.5.4 Position Error. The standard transformer shall be adjusted to provide a voltage across pins A and C corresponding to pressure specified in the detail specification. The reading of the indicator, taken while the indicator is in a normal upright position, shall not differ from the reading while the indicator is held in any other position by more than the amount specified in the detail specification.

4.5.5 Sealing. A mass spectrometer type of helium leak detector shall be used for leak tests. The leak rate shall not exceed 0.1 micron cubic foot per hour.

4.5.6 Vibration. Vibration testing shall be accomplished in accordance with the requirements specified in the detail specification.

4.5.7 Low Temperature. The indicator shall be subjected to a temperature of $-54^{\circ} \pm 2^{\circ}C$ for a period of 4 hours. While at this temperature, the indicator shall be tested for scale error and friction as specified in 4.5.3. Scale error and friction tolerances shall not exceed values specified in the detail specification.

4.5.8 High Temperature. The indicator shall be subjected to a temperature of $71^{\circ} \pm 2^{\circ}C$ for a period of 4 hours. While at this temperature, the indicator shall be tested for scale error and friction as specified in 4.5.3. Scale error and friction tolerances shall not exceed values specified in the detail specification.

4.5.9 Fogging. The indicator shall be placed in a controlled ambient temperature of $85^{\circ} \pm 5^{\circ}C$ for a minimum of 2 hours. Within 10 seconds after removal from the heated environment, the face of the indicator shall be immersed to a

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depth of 1/4 inch in an ice water bath maintained at 0° to 7°C. After 3 minutes, the indicator shall be removed from the bath and quickly wiped dry (compressed air shall not be used.) Cover glass and dial shall immediately be inspected. Any evidence of fogging or damage shall be cause for rejection.

4.5.10 Electrical Characteristics. The electrical characteristics of the indicator shall be checked to determine compliance with 3.5.6 through 3.5.6.5.

4.5.11 Magnetic Effect. The indicator shall be rotated in a vertical plane about a short bar magnetic compass with the nearest part of the indicator 5 1/2 inches from, and magnetically East or West of the center of the compass, the indicator shall be held in positions, 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in a normal operating position. This procedure shall be done twice. First, not operating and second, operating on rated voltage. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. Deflection of compass at any of the specified positions shall not exceed 1°.

4.5.12 Effect of Lead Length. The indicator shall not change more than 1 psi when the lead length between the indicator and transmitter is changed from 5 to 300 feet of number 20 AWG gage wire. Equivalent resistance may be substituted for added wire length.

4.5.13 Temperature-Altitude. The temperature altitude test shall be conducted in accordance with MIL-STD-810, Method 504, procedure I (except temperature shall be -55°C), with a chamber pressure equivalent to an altitude of 50,000 feet. Rate of pressure change in the chamber shall not exceed an average of 3 psi per second while under these conditions, and at the end of a 1 hour period the scale error and friction test shall be conducted. Errors shall not exceed tolerances specified in the detail specification. This test may be conducted concurrently with the low temperature exposure test.

4.5.14 Low Temperature Exposure. The indicator shall be subjected to a temperature of -65° ± 2°C for a period of 48 hours. While at this temperature, the indicator shall be operated to assure that the pointer moves over the entire scale range without exhibiting sticky or erratic characteristics. Temperature shall then be raised to -55° ± 2°C and held for a period of 24 hours. While at this temperature, the indicator shall be subjected to scale error and friction test and shall not require more than 5 seconds to come within tolerances specified in the detail specification. The indicator shall then be maintained at room temperature for at least 1 hour and then subjected to scale error test at room temperature. No damage or deterioration shall have occurred and scale errors shall not exceed those specified in the detail specification for room temperatures.

4.5.15 High Temperature Exposure. The indicator shall be subjected to a temperature of 70° ± 2°C for a period of 24 hours. The indicator shall then be tested at this temperature for scale errors as specified in para 4.5.3.

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Scale error and friction tolerances shall not exceed values specified in the detail specification. The indicator shall then be maintained at room temperature for at least 1 hour, and then subjected to scale error test at room temperature. No damage or deterioration shall have occurred and scale errors shall not exceed those specified in the detail specification for room temperature.

4.5.16 Thermal Shock. The indicator shall be subjected to eight continuous cycles of immersion in water at a temperature of $85^{\circ} \pm 2^{\circ}C$, and then in water at a temperature of $5^{\circ} \pm 2^{\circ}C$ without evidence of moisture penetration or damage to the coating of the case. The duration of each immersion shall be 30 minutes, with not more than 5 seconds between immersions. After completion of the eight specified cycles, the indicator shall be subjected to the test specified in para 4.5.5. No leakage shall occur.

4.5.17 Salt Fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509, procedure I, for a period of 48 hours and then examined. There shall be no evidence of external corrosion or deterioration that would affect subsequent operation. The indicator shall be subjected to the test specified in para 4.5.2. Measured resistance shall be a minimum of 20 megohms. Indicator shall not exceed tolerances specified in the detail specification.

4.5.18 Damping. With excitation power off, test transmitter shall be positioned to the equivalent of 60% of full scale. Excitation power shall then be applied and the time for the indicator to show 60% of full scale shall not exceed 2.0 seconds.

4.5.19 Pointer Smoothness. The indicator shall be driven from a suitable matched source such as a transmitter conforming to MIL-T-58097. The input signal to the indicator shall be varied at a constant rate so as to cause the indicator pointer to move from zero scale to full scale to full scale and back to zero scale in a period of 1 minute $\pm$ 10 seconds. Two complete cycles shall be performed while observing the indicator pointer. The pointer shall track smoothly over the full scale range. Any erratic motion of the pointer or evidence of sticking shall be cause for rejection. The instrument shall not be tapped or vibrated at any time during this test.

4.5.20 Shock. The indicator shall be subjected to a shock test in accordance with MIL-STD-810, Method 516, procedure I, Figure 516-1; as prescribed for flight vehicle equipment.

4.6 Manufacturing Run-in Test. Ten indicators shall be selected at random from every 100 or less indicators produced. The indicators shall be mounted on a vibration table in the positions indicated in Table II and then subjected to the following tests:

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TABLE II

NUMBER OF INDICATORS	PITCH POSITION: DEGREES TOP OF DIAL IS TILTED REARWARD FROM 0 DEGREE	ROLL POSITION: DEGREES ROTATED FROM NORMAL OPERATING POSITION
2	20	NORMAL POSITION
1	25	25° CCW
1	25	25° CW
1	20	45° CCW
1	20	24° CW
1	15	60° CCW
1	15	60° CW
1	15	90° CCW
1	25	90° CW

4.6.1 Driving Source Stopped. All ten indicators shall be connected to a driving source equivalent to that specified in para 4.5.19 and the driving source energized. The driving source shall be stopped with the indicators reading approximately mid scale, then half the units shall be vibrated in the lateral axis and half the units vibrated in the longitudinal axis by a 2.2 G ± 10% sinusodial force at a frequency between 20 and 60 HZ for a period of 30 minutes. The units shall be monitored for pointer oscillation, which may not exceed ± 0.5 psi.

4.6.2 Driving Source Energized. The driving source shall then be energized for an additional 30 minutes with vibration force still applied. During this period the units shall be continuously monitored for pointer smoothness. Any erratic motion or evidence of sticking by the pointer shall be cause for rejection.

4.6.3 Vibration Stopped. The vibration force shall be stopped while driving force is continued for 30 additional minutes while monitoring the units for pointer operation. Any evidence of erratic motion or pointer sticking shall be cause for rejection.

4.6.4 Duration. The above procedures shall be repeated 4 times, accumulating a total of at least 6 hours of test time. All ten units shall then be subjected to the scale error and friction tests specified in para 4.5.3.

4.6.5 Rejection and Retest. If any unit fails to pass any requirement of the manufacturing run-in test, then 100% testing is mandatory until a 2% or less rejection is established.

5. PACKAGING

5.1 Preservation, Packing, and Marking. Preservation, packing, and marking of indicators shall be as specified in MIL-E-17555.

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6. NOTES

6.1 Intended Use. The indicators covered by this specification are intended for use with voltage ratio transmitters conforming to MIL-T-58097 in providing remote indication of aircraft pressure functions.

6.2 Ordering Data. Ordering data shall be as specified in the detail specification.

6.3 Definitions.

6.3.1 Failure. Where a condition is specified as a requirement, a failure will be considered as any condition which fails to comply with the requirements specified herein.

6.4 Reclaimed Materials. This specification requires the use of recovered materials to the maximum extent possible without jeopardizing the intended end use of the item (see 3.3.7).

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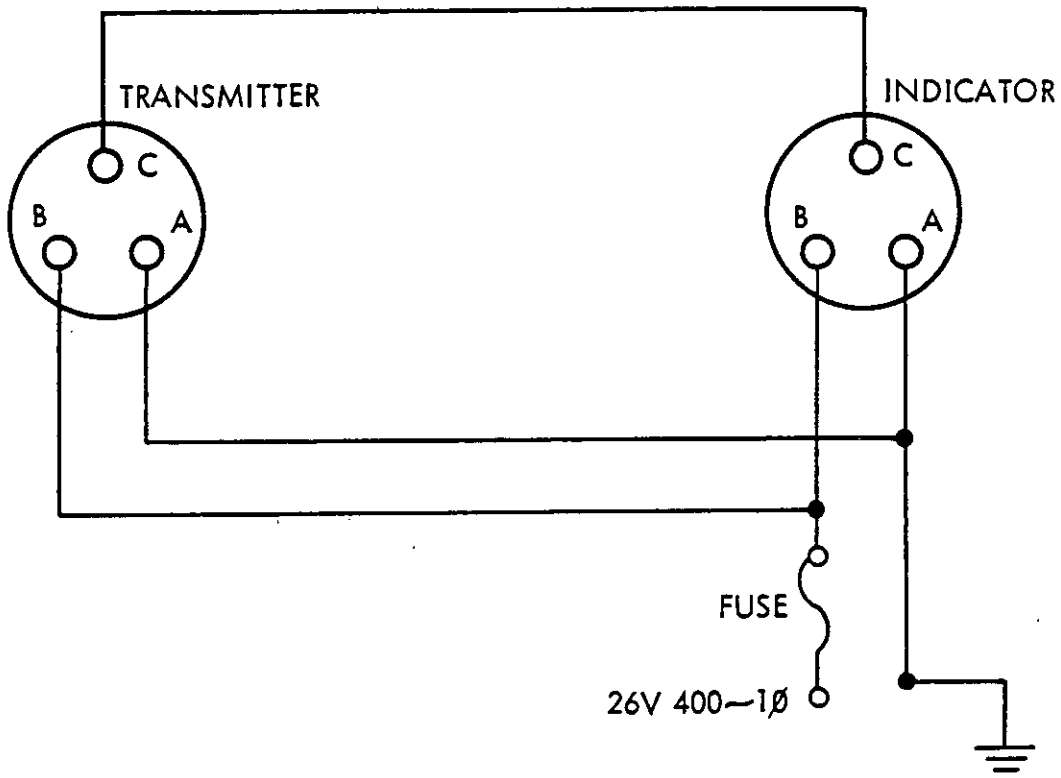
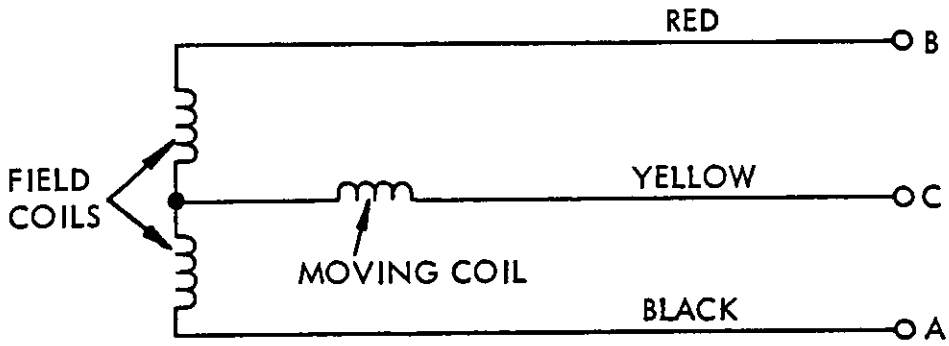


Figure 1. External Wiring Diagram



A-B- POWER 26V 400—1Ø
A-C- TAP VOLTAGE

FIGURE 1. INTERNAL WIRING DIAGRAM

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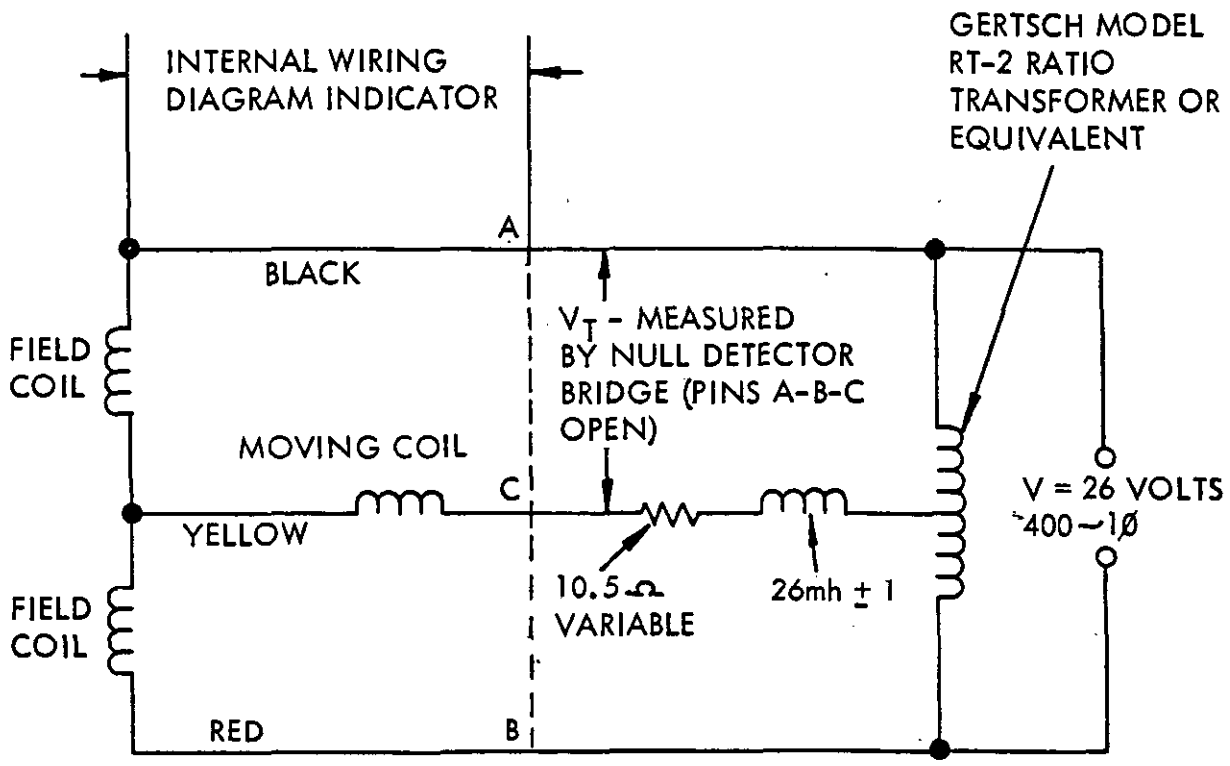
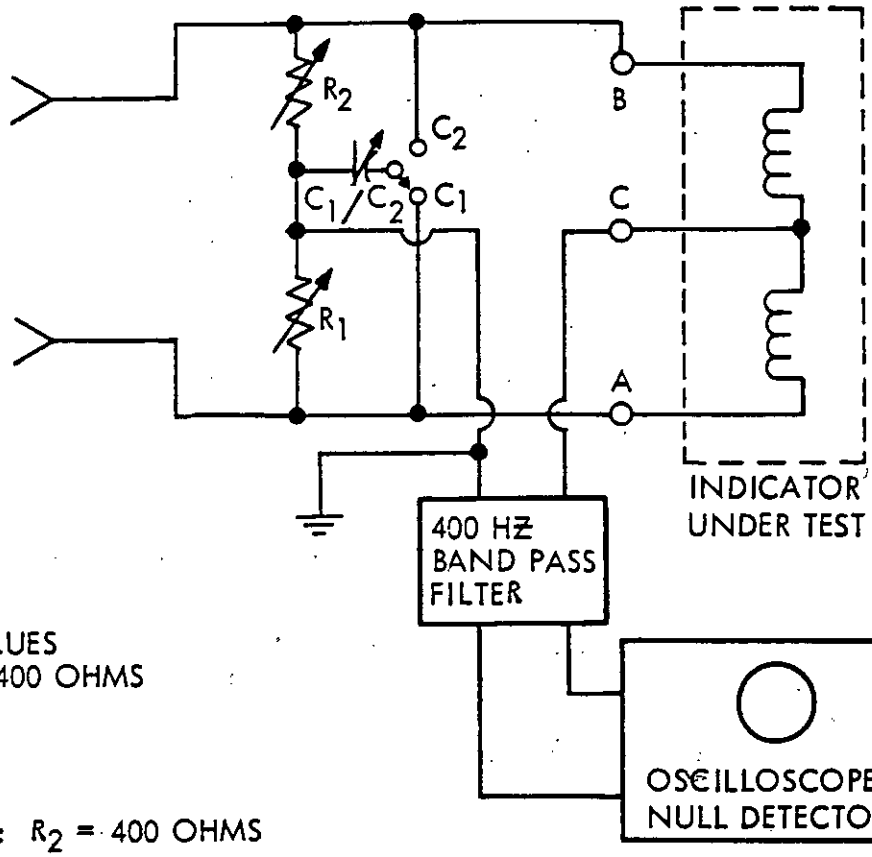


FIGURE 2. TEST STANDARD CIRCUIT

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SUGGESTED TEST CIRCUIT VALUES
FOR ZERO DEFLECTION: R_1 = 400 OHMS
NULL WITH R_2 AND C_2
$R_2 \approx$ 300 OHMS
$C_2 \approx$ 0.1000 UFD

FOR FULL SCALE DEFLFCTION: R_2 = 400 OHMS
NULL WITH R_1 AND C_1
$R_1 \approx$ 300 OHMS
$C_1 \approx$ 0.1200 UFD

$$X = \frac{R_1 (R_1 + R_2 + R_1 \times R_2^2 \times \omega^2 \times C_2^2)}{(R_1 + R_2)^2 + (R_1 \times R_2 \times \omega)^2 (C_1^2 + C_2^2)}$$

$$Y = \frac{R_1 \times R_2 \times \omega \times (R_2 \times C_2 - R_1 \times C_1)}{(R_1 + R_2)^2 + (R_1 \times R_2 \times \omega)^2 (C_1^2 + C_2^2)}$$

TAP VOLTAGE RATIO = E = $\sqrt{X^2 + Y^2}$

ϕL = ARC TAN $\frac{Y}{X}$

FIGURE 3. TAP VOLTAGE MEASURING BRIDGE

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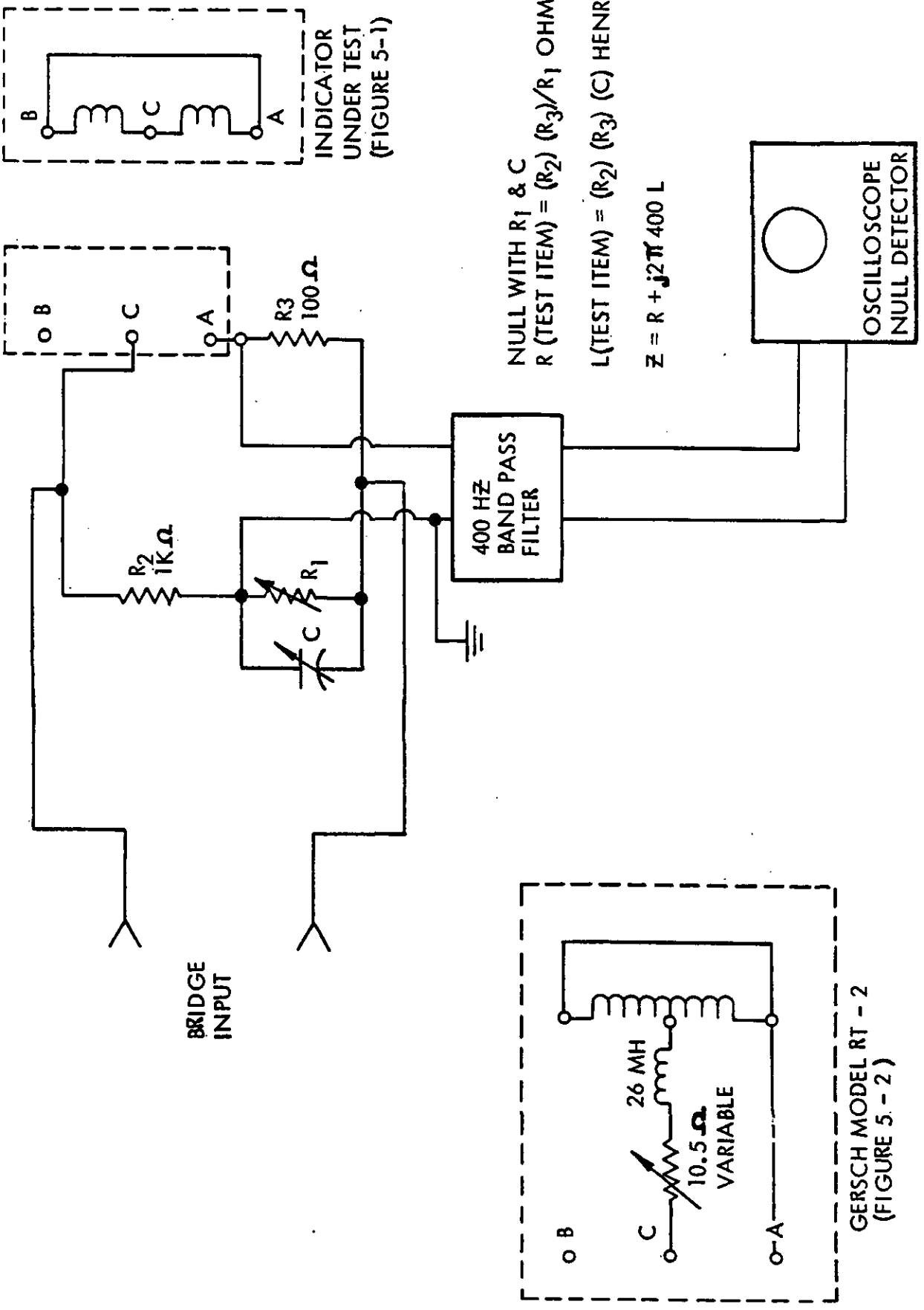


FIGURE 4. THEVENIN IMPEDANCE MEASURING BRIDGE

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